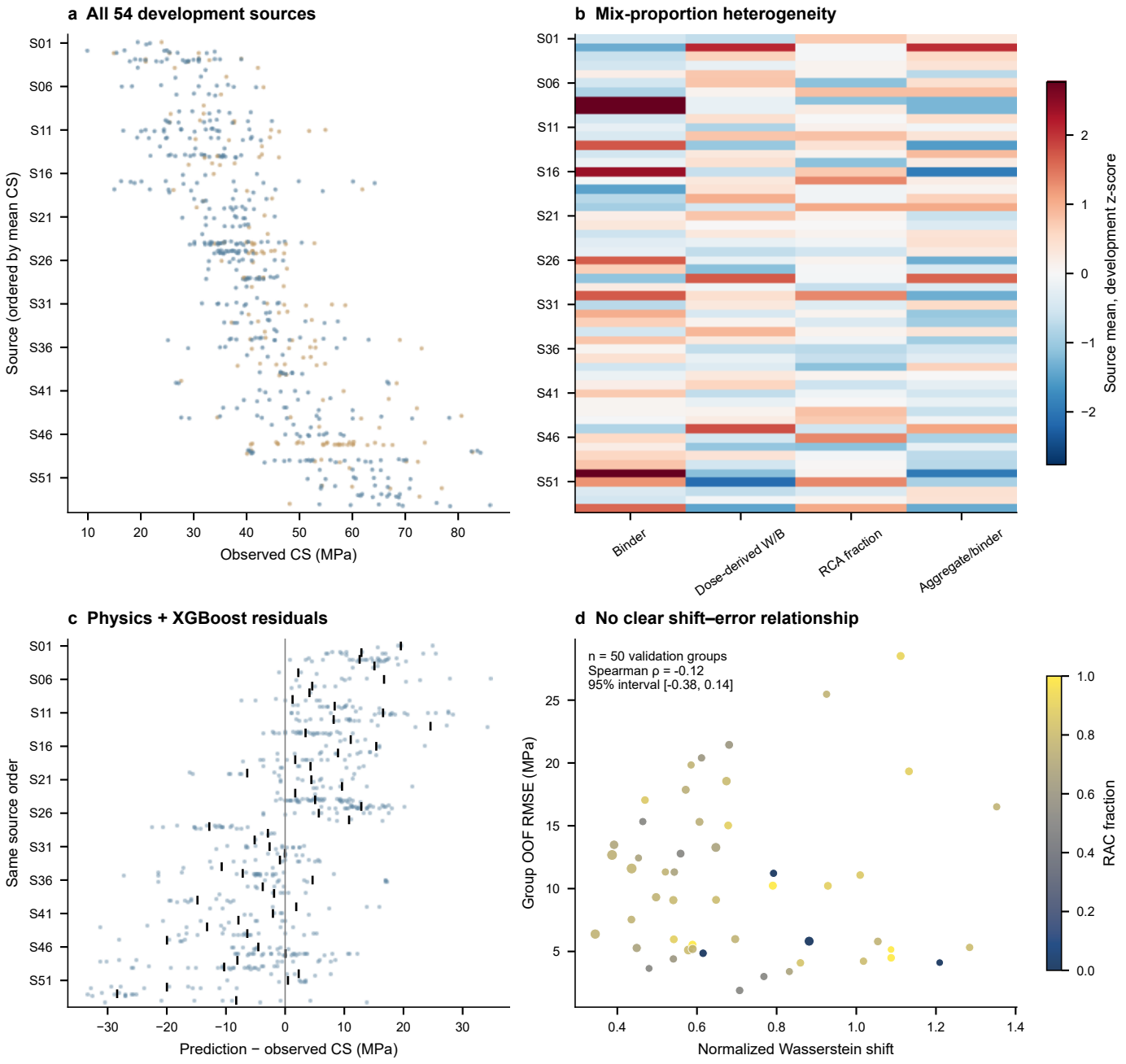

a All 54 development sources
Source (ordered by mean CS)
S01
S06
S11
S16
S21
S26
S31
S36
S41
S46
S51
10
20
30
40
50
60
70
80
Observed CS (MPa)
b Mix-proportion heterogeneity
Binder
Dose-derived W/B
RCA fraction
Aggregate/binder
Source mean, development z-score
2
1
0
−1
−2
c Physics + XGBoost residuals
Same source order
−30
−20
−10
0
10
20
30
Prediction − observed CS (MPa)
d No clear shift–error relationship
n = 50 validation groups
Spearman ρ = -0.12
95% interval [-0.38, 0.14]
Group OOF RMSE (MPa)
5
10
15
20
25
0.4
0.6
0.8
1.0
1.2
1.4
Normalized Wasserstein shift
RAC fraction
0.0
0.2
0.4
0.6
0.8
1.0